

TABLE 4
Study Variables and Summary Statistics (*continued*)

Variable	Coding	Mean	Std. Dev.
Hours per week of indoor recreation	Continuous	2.95	5.89
Hours per week of outdoor recreation	Continuous	4.95	7.11
Has a regular doctor	1 = yes, 0 = no	0.89	0.31
<i>Demographics</i>			
Male	1 = male, 0 = female	0.60	0.49
Married	1 = yes, 0 = no	0.69	0.46
Age	Continuous	59.11	15.37
White	1 = yes, 0 = no	0.79	0.41
Graduate school graduate	1 = yes, 0 = no	0.20	0.40
College/technical school graduate	1 = yes, 0 = no	0.62	0.49
Employed full-time	1 = yes, 0 = no	0.48	0.50
Employed part-time	1 = yes, 0 = no	0.08	0.27
Not employed	1 = yes, 0 = no	0.42	0.49
Has health insurance	1 = yes, 0 = no	0.92	0.27
Number of children under 18 years old in household	Continuous	0.43	0.83
Lives in Duarte	1 = yes, 0 = no	0.13	0.34
Lives in Monrovia	1 = yes, 0 = no	0.20	0.40
Lives in Sierra Madre	1 = yes, 0 = no	0.08	0.26
Lives in Burbank	1 = yes, 0 = no	0.19	0.40
Lives in Glendora	1 = yes, 0 = no	0.40	0.49
Income	15 = < 19,999; 25 = 20,000–29,999; 35 = 30,000–39,999; 45 = 40,000–49,999; 55 = 50,000–59,999; 65 = 60,000–69,999; 75 = 70,000–79,999; 85 = 80,000–89,999; 95 = 90,000–99,999; 125 = 100,000–149,999; 175 = 150,000–199,999; 200 = > 200,000	83.52	53.50
<i>Beliefs</i>			
Heard or read about possible health effects	1 = yes, 0 = no	0.86	0.35
Believes smoke can affect health	1 = yes, 0 = no	0.90	0.31

TABLE 5
“Yes” Responses by Bid Amount (*N* = 157)

Bid Amount	N	Yes (%)
\$10	22	59
\$25	21	67
\$50	18	44
\$75	18	11
\$100	14	50
\$150	12	25
\$200	7	29
\$300	11	18
\$500	19	11
\$750	15	13

address this issue of endogeneity must be employed. To test for endogeneity of the defensive action variables, we use a version of the Hausman specification error test (see Hausman 1976; Gujarati 2003).² Results indicate

² Preliminary analysis shows that only three defensive actions, “Used a home air cleaner,” “Ran air conditioner more than usual,” and “Avoided normal outdoor recreation/exercise,” could be explained by an appropriate set of instrumental variables, which is a required feature to employ this test. These instrumental variables include “Months at current zip code,” “Income,” “Employed full-time,” and “Believes smoke can affect health.”

that “Used a home air cleaner” is the only defensive action variable that has a negative and statistically significant effect on the expected number of symptom days, and is also endogenous. Air cleaners and purifiers are recommended in the home during wildfires to help reduce indoor particle levels (Lipsett et al. 2008; USEPA 2010a). Henderson, Milford, and Miller (2005) conducted a study on the effectiveness of air cleaners during wildfires and prescribed burns and found that homes with air cleaners experienced 63% to 88% less particulate matter in the home than those without air cleaners. Mott et al. (2002) also found that greater use of high-efficiency air cleaners in the home was associated with reduced odds of reporting adverse health effects during a 1999 wildfire.

To estimate the health production function and address the issue of endogeneity in a non-linear framework, we use a maximum simulated likelihood estimation procedure developed by Deb and Trivedi (2006a, 2006b).³ The model has the following equations for the count dependent variable “Symptom days” and the binary endogenous regressor “Used a home air cleaner”:

$$\Pr[Y_i = y_i | \mathbf{x}_i, d_i, l_i] = f(\mathbf{x}_i' \beta + \gamma d_i + \lambda l_i), \quad [16]$$

$$\Pr[d_i = 1 | \mathbf{z}_i, l_i] = g(\mathbf{z}_i' \alpha + \delta l_i). \quad [17]$$

In the health outcome equation [16], y_i represents the total number of days that symptoms from exposure to the wildfire smoke were experienced, and $\mathbf{x}_i$ represents a vector of exogenous variables that could affect the number of symptom days experienced, such as objective or subjective pollution levels, the type of symptom experienced, health history, demographics, and lifestyle factors, with associated parameters β . The binary endogenous regressor, “Used a home air cleaner,” is represented by d_i , with associated parameter γ . Finally, l_i represents latent factors that can affect the decision to use a home air cleaner, as well as the number of symptom days experienced, with associated parameters λ . These latent factors capture the unobserved

heterogeneity that is the cause of endogeneity in the model.

“Used a home air cleaner” is modeled in equation [17] where $\mathbf{z}_i$ represents a vector of exogenous variables that could affect the probability that a home air cleaner is used, with associated parameters α . These could be pollution levels, the type of symptom experienced, health history, demographics, lifestyle factors, as well as beliefs about the effects of wildfire smoke on health. Again, this equation includes latent factors l_i with associated parameters δ .

Equations [16] and [17] can contain the same set of exogenous variables; however, for more robust identification, instrumental variables that are included in the equation for “Used a home air cleaner” but excluded from the equation for “Symptom days” can be used. The error term in each equation is partitioned into latent factors l_i and an independently distributed random error term. The joint distribution of “Symptom days” and “Used a home air cleaner” can then be specified, conditional on these common latent factors. While the latent factors are unknown, it is assumed that they come from a normal distribution, allowing them to be integrated out of the joint density. These equations are then estimated jointly by simulation-based estimation. “Symptom days” is modeled with a negative binomial distribution, given the presence of overdispersion in the data, and “Used a home air cleaner” is assumed to follow a normal distribution. Two thousand simulation draws are used based on recommendations from Deb and Trivedi (2006a), and robust standard errors that take simulation error into account are reported. The results of this model, including only those variables that had a statistically significant effect on expected symptom days, can be found in Table 6.⁴

As expected, perceived pollution levels as measured by whether smoke was smelled both inside and outside the home for greater than five days has a positive and statistically significant effect on the expected number of

³ We graciously thank Partha Deb for providing access to his Stata program *treatreg2* to estimate this model.

⁴ The results of the full model are available from the authors.

TABLE 6
Defensive Behavior Model

Variable	Coef.	Robust Std. Err.
<i>Symptom Days, Negative Binomial Regression</i>		
Smelled smoke indoors > 5 days	0.394***	0.142
Smelled smoke outdoors > 5 days	0.953***	0.168
Ear, nose or throat symptoms	3.630***	0.232
Breathing symptoms	0.789***	0.183
Other symptoms	0.719***	0.221
Used a home air cleaner	-0.848***	0.163
Hours per week of outdoor recreation	-0.023*	0.012
Male	-0.341**	0.151
Married	-0.345**	0.153
Age	0.012**	0.005
College/technical school graduate	0.479***	0.141
Employed part-time	0.625**	0.305
Lives in Duarte	0.539**	0.225
Lives in Burbank	0.460**	0.185
Lives in Glendora	0.406**	0.174
Constant	-3.701***	0.476
<i>Used a Home Air Cleaner, Probit Regression</i>		
Smelled smoke indoors > 5 days	0.362	0.259
Smelled smoke outdoors > 5 days	0.336	0.282
Ear, nose or throat symptoms	0.672***	0.242
Breathing symptoms	0.168	0.265
Other symptoms	1.374***	0.333
Hours per week of outdoor recreation	-0.017	0.021
Male	-0.183	0.246
Married	0.437	0.268
Age	-0.006	0.010
College/technical school graduate	0.375	0.248
Employed full-time	0.560**	0.284
Employed part-time	0.519	0.461
Lives in Duarte	-0.220	0.400
Lives in Burbank	0.411	0.307
Lives in Glendora	0.496*	0.272
Income	-0.005**	0.003
Believes smoke can affect health	1.426**	0.703
Constant	-3.481***	1.096
/ln α	0.858***	0.072
N	-13.657***	2.491
Log likelihood	377	
Wald χ^2 (24)	-672.066	
Prob > χ^2 =	424.71	
	0.0000001	

* $p \leq 0.10$; ** $p \leq 0.05$; *** $p \leq 0.01$.

symptom days experienced, all else constant. Similarly, based on a questionnaire of high school students and children during the 2003 California wildfires, Kunzli et al. (2006) found that the risk of experiencing all health symptoms increased monotonically with perceived pollution levels, as measured by the number of reported smoky days. In addition, we find that having ear, nose, or throat symp-

toms, breathing symptoms, or other symptoms during the wildfire has a positive and statistically significant effect on the expected number of symptom days, compared to having heart symptoms. Using a home air cleaner is the one defensive action that is found to have a negative and statistically significant effect on expected symptom days. This finding is consistent with prior research (Mott et al.

2002; Henderson, Milford, and Miller 2005). Being a male or married has a negative effect on expected symptom days, compared to individuals who are female or single. Expected symptom days increase with age, which is not surprising given warnings by the CDC and EPA that older individuals are at greater risk of experiencing health effects from exposure to wildfire smoke.

Various factors also have an influence on whether an air cleaner was used in the home to reduce exposure to the smoke from the Station Fire. The positive and significant coefficient on the latent factor, lambda, suggests that individuals who are more likely to use an air cleaner, based on unobserved characteristics, are more likely to experience symptom days. This could reflect some predisposition to getting sick. For instance, individuals who are more likely to experience symptoms from wildfire smoke exposure may realize this, and as a result they may be more likely to take defensive actions, such as using an air cleaner in their home during a wildfire.

Contingent Valuation Model

In applying a CVM framework to value a decrease in the number of symptom days experienced from exposure to a pollutant, WTP will be a function of the bid amount and any variables that would enter the health production function. The CVM portion of the survey questioned respondents about whether any members of their household experienced health symptoms from the smoke from the Station Fire. If they indicated that they had, the respondents were asked if they would be willing to pay a specified bid amount to reduce the number of symptom days experienced by any member of the household by 50%.

Ultimately, we would like to know the actual WTP distribution of all respondents, but given the dichotomous choice question format used here, the only known information is whether a respondent responded "yes" to a specified bid amount, in which case that respondent's actual WTP is greater than or equal to this value, or responded "no," in which case the actual WTP is less than this value. Thus, the actual underlying WTP distribution of in-

terest, which we refer to as WTP^* , is unknown. Following closely the work of Alberini (1995), a linear WTP model can be specified as

$$WTP_i^* = \mathbf{x}_i' \beta + \varepsilon_i, \quad [18]$$

where $\mathbf{x}_i$ represents a vector of independent variables that could influence the individual's WTP, and ε is a normally distributed error term. Whether or not an individual was willing to pay a specified bid amount is observed, so the probability that the individual responds "yes" to a specified bid amount " bid_i " is equal to the probability that the random WTP function is greater than that offered bid amount:

$$Pr(WTP_i^* \geq bid_i | \mathbf{x}_i') = 1 - F(bid_i | \mathbf{x}_i'), \quad [19]$$

where F is the cumulative distribution function of WTP_i^* . This model is estimated by the method of maximum likelihood, which requires that a distribution is specified for the underlying WTP distribution. Various probability distributions were considered to model WTP; however, a log-normal functional form is chosen for two main reasons. First, while a logistic distribution is frequently assumed, it has been noted that a logit regression model should have a sample size of at least 500 observations (Studenmund 1992). However, in this study, only 157 respondents were eligible to respond to the CVM question. Giraud, Loomis, and Cooper (2001) cite less need for a large sample size as an advantage of the probit model over the logit model in estimating WTP values. Second, assuming a nonnegative distribution for WTP seems reasonable for the case of valuing a decrease in symptom days from exposure to wildfire smoke. It seems implausible that individuals would hold negative values for a decrease in a health outcome whose presence is expected to reduce their utility. As Alberini and Cooper (2000) point out, a negative WTP value would indicate that the average individual would actually pay to be sick. The log of the bid amount is included in the model to restrict WTP values to lie between zero and infinity. Assuming WTP follows this log-normal distribution, the WTP model can be specified as